SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO4:** Articulation (BL4, BL5)	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	**Total Marks:**	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	**Ruthika Reddy**	**Reg. No.:**	**192421138**
Section / Batch:	**Btech IT**	**Date:**	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

SIMATS ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

C RUTHIKA REDDY
192421138RA

QUESTIONS

Q1
Smart Retail Checkout Optimization (Queue + Scheduling Algorithms)
100 marks

1/1 answered

Q1 **Smart Retail Checkout Optimization (Queue + Scheduling Algorithms)** 100 marks

A retail store must manage checkout counters to reduce customer waiting time. Analyze how queueing and scheduling algorithms can optimize service. Identify constraints such as peak hours and staff availability. Design a solution to balance load across counters. Discuss scalability in large retail chains. Evaluate performance metrics. Interpret results in improving customer experience.

Your Code

```
import heapq
import random
class CheckoutSimulation:
    def __init__(self, num_counters):
        self.num_counters = num_counters
        self.counters = [[0, i] for i in r
        heapq.heapify(self.counters)
        self.customer_log = []
    def assign_customer(self, customer_id,
        least_loaded = heapq.heappop(self.
        wait_time = least_loaded[0]
        least_loaded[0] += service_time
        heapq.heappush(self.counters, leas
        self.customer_log.append({
            "customer": customer_id,
```

```
                "counter": least_loaded[1],
                "wait_time": wait_time,
                "service_time": service_time
            })
    def run_simulation(self, num_customers
        for cust_id in range(1, num_custor
            service_time = random.randint
            self.assign_customer(cust_id,

    def report(self):
        total_wait = sum(c["wait_time"] fo
        avg_wait = total_wait / len(self.c
        max_load = max(c[0] for c in self.

        print("---- Checkout Simulation Re
        for c in self.customer_log:
            print(f"Customer {c['customer'
                  f"| Wait: {c['wait_time'

        print("\n---- Performance Metrics
        print(f"Average Waiting Time : {av
        print(f"Max Counter Load     : {ma
        print(f"Counters Used        : {se


if __name__ == "__main__":
    num_counters = 4
    num_customers = 15

    sim = CheckoutSimulation(num_counters)
    sim.run_simulation(num_customers)
    sim.report()
```

Input

```
stdin input
```

Output

```
---- Checkout Simulation Report ----
Customer   1 -> Counter 0 | Wait:   0 min
| Service: 5 min
Customer   2 -> Counter 1 | Wait:   0 min
| Service: 3 min
Customer   3 -> Counter 2 | Wait:   0 min
```

Submitted